LAB 1: Error Analysis and Orientation

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Abstract *This lab report examines the principles of error propagation and uncertainty. The length and width of a rectangle is measured consistently as it moves over the table, using the tracking camera. The measurement data received by the tracking camera is then used to calculate the average length, average width, and their individual uncertainties. Using these calculations, the total area of the rectangle is computed, as well as its uncertainty using error propagation. The total area is also calculated for each reading and then a statistical average and uncertainty is computed and the two total area results are compared. The uncertainty associated with the tracking camera is assessed using the areas and uncertainties computed during this procedure.*

Keywords: *area, camera, uncertainty, histogram*

1. Introduction

The purpose of this lab was to obtain a general idea of the uncertainty associated with the tracking camera. In order to achieve this, the tracking camera took the length and width of a rectangle repeatedly, and these measurements were then used to calculate the total area and its uncertainty in two ways. Using the physics concept of error propagation and statistics, the uncertainty of the tracking camera could be analyzed.

The area of the rectangle was calculated using the area of rectangle formula:

$$A = l \times w \qquad \textbf{\textit{Equation 1}}$$

Then average length and width were calculated using the sample mean formula:

$$\bar{x} = \frac{x_1 + x_2 + \ldots + x_n}{n} \qquad \textbf{\textit{Equation 2}}$$

Then the standard deviation was calculated using the sample standard deviation formula:

$$\sigma = \sqrt{\frac{1}{(n-1)} \sum_{i=1}^{n} (x_i - \bar{x})^2} \qquad \textbf{\textit{Equation 3}}$$

Then the statistical uncertainty was calculated using the standard error formula:

$$\delta\bar{x} = \frac{\sigma}{\sqrt{n}} \qquad \textbf{\textit{Equation 4}}$$

Then the propagated uncertainty was calculated using the error of propagation for multiplication formula:

$$\delta f(w, l) = \sqrt{(l\,\delta w)^2 + (w\,\delta l)^2} \qquad \textbf{\textit{Equation 5}}$$

2. Experimental Procedure

In order to compute the dimensions of the rectangle we use a compute vision system to analyze the positions of colored markers, called "dots" derived from their circular shape. We place dots at three of the four corners, each with a distinct color. Our particular setup, shown in **(figure 1)**, includes a green dot on the top-left corner of the rectangle, an orange dot on the top-right corner of the rectangle, and a hot pink dot on the bottom-right corner of the rectangle. We define dimension "width" to be the center-to-center distance between the green and orange dots, and dimension "length" to be the center-to-center distance between the orange and hot pink dots.

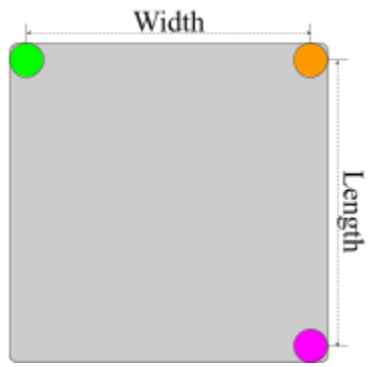

Figure 1: Setup of Rectangle

The rectangle sits face up on a low-friction surface. A camera is positioned above the surface, facing down, such that the markers on the rectangle are visible in the frame. During runtime, the computer vision tool takes frames at even time intervals, then identifies where the markers are by filtering for the given color. For each of these intervals, the coordinate of the center pixel relative to the origin of the frame is appended to a CSV log file. We ran the tool to log 2500 distinct samples, moving the rectangle around the surface. The results were then calculated using the methods described in the Results and Analysis section.

3. Results and Analysis

The first step in the analysis process involved culling the data to remove data where position data is missing from one of the colored dots. This was performed by manually removing incomplete rows from the spreadsheet software. The raw data includes x and y position information about each colored dot at each point in time. For each sample, we can compute a width and length using the distance formula. The distance between the dark orange and green dot was chosen to represent the width, while the distance between the dark orange and hot pink dot was the length. Using these pixel figures, we can compute an area for each sample in pixel^2 (**equation 1**). A truncated table of the data and resulting row wise computations are shown in (**table 1**) to illustrate.

Table 1: Truncated Sample Data and Row Wise Computations

position_px_x-darkorange	*position_px_y-darkorange*	*position_px_x-green*	*position_px_y-green*	*position_px_x-hotpink*	*position_px_y-hotpink*	*Width (pixels)*	*Length (pixels)*	*Area (pixels^2)*
698	784	612	637	555	871	170.3085435	167.385782	28507.22873
…	…	…	…	…	…	…	…	…

The spreadsheet software was used to create (**figure 2**) and (**figure 3**) which are histograms displaying the distribution of widths and lengths in the data respectively.

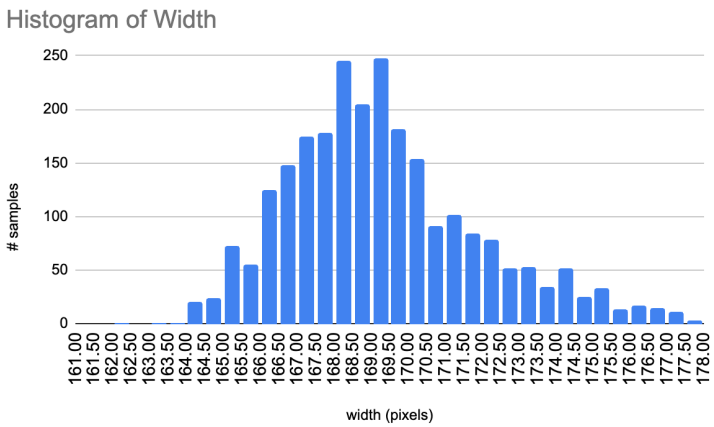

Figure 2: Distribution of Width

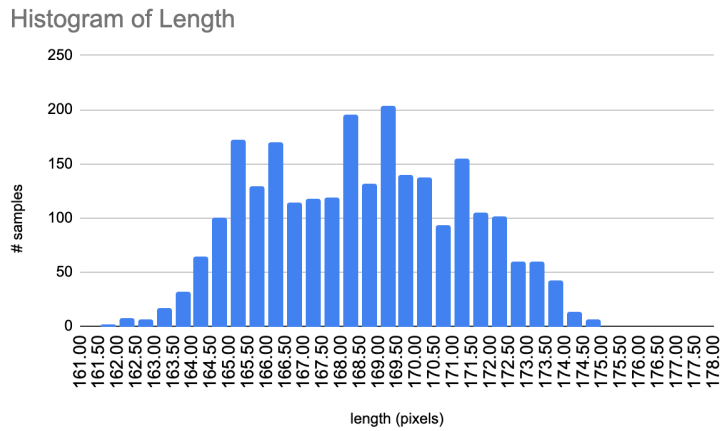

Figure 3: Distribution of Length

With all the row wise data tabulated, it is now possible to calculate statistics that describe our data as a whole. The numerical results of the following calculations can be found in (**table 2**). All calculation results are in pixels unless mentioned otherwise. The first step is to calculate the mean length and width from all of the samples (**equation 2**). This allows us to produce the standard deviation of the samples for both length and width (**equation 3**). Then, we can calculate the direct statistical uncertainty for the length and width using the standard error formula (**equation 4**). Finally, we can use the same equations to compute standard deviation and statistical uncertainty of the areas column in pixels^2 (**equations 3,4**).

With all of this information, we can assess the area of the rectangle with associated uncertainties in two different ways. Numerical results of these calculations are shown in (**table 2**). Method 1 involves multiplying the mean length/width figures (**equation 1**) and propagating the uncertainties using (**equation 5**) to produce a final area figure. Method 2 directly computes the mean of the areas column (**equation 2**) and uses the statistical uncertainty of the areas column as a final error figure (**table 2**).

Table 2: Means, Standard Deviations, and Error Figures

Value	**Measurement**	**Standard Deviation**	**Uncertainty**
Mean length pixels	168.51	2.6808	0.053616
Mean width pixels	169.31	2.6154	0.052308
Area pixels^2 (Method 1, propagated)	28529	N/A	12.653
Area pixels^2 (Method 2, statistical)	28532	742.000	14.840

4. Conclusions

In the aforementioned lab we analyze a rectangle's dimensions using an optical tracking system. By collecting a large set of data points on the relative positions of optical markers strategically positioned on the rectangle, we are able to compute the expected length, width, area, and associated uncertainties, including calculating both the propagated and statistical uncertainty for area. In addition, we present visual representations of the measurement distributions to further qualify this uncertainty. As a result, we are able to effectively analyze the uncertainty in the measurement from the optical tracking system.